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Section:6

Skill Experiment -1

Git Version Control

Introduction to Git

Git is a distributed version control system that helps you track changes in your codebase. It allows multiple people to collaborate on a project without overwriting each other's work. Git keeps a complete history of changes and enables you to revert to previous versions, compare code, and merge different versions of files.

Basic Git Commands

Initializing a Git Repository

To start using Git, navigate to your project directory and initialize a Git repository.

```
cd /path/to/your/project
 git init
```

Checking the Status

To see the current state of your working directory, including changes not yet staged for commit:

```
 git status
```

Adding Files to Staging

Before committing changes, add files to the staging area:

```
 git add <file-name>      # Add a specific file
 git add .                # Add all files
```

Committing Changes

After staging your changes, commit them with a descriptive message:

```
 git commit -m "Your commit message here"
```

Viewing Commit History

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To view the commit history of your project:

git log

Creating a Branch

Branches allow you to work on different features or bug fixes without affecting the main codebase.To create a new branch:

git checkout -b

Switching Branches

To switch to an existing branch:

git checkout<branch-name>

Merging Branches

Once you've made changes on a branch, you may want to merge it into the main branch (or another branch). First, switch to the branch you want to merge into (e.g., main):

git checkout main

Then merge the changes:

git merge<branch-name>

Deleting a Branch

After a branch has been merged, you can delete it:

git branch -d<branch-name>

Collaboration with Git

Pulling Changes from a Remote Repository

To fetch and merge changes from a remote repository:

git pull origin main

Resolving Merge Conflicts

If there are conflicting changes between branches, Git will notify you, and you'll need to manually resolve the conflicts by editing the affected files. After resolving, mark them as resolved:

git add<branch-name>

git commit -m "Resolved merge conflict"

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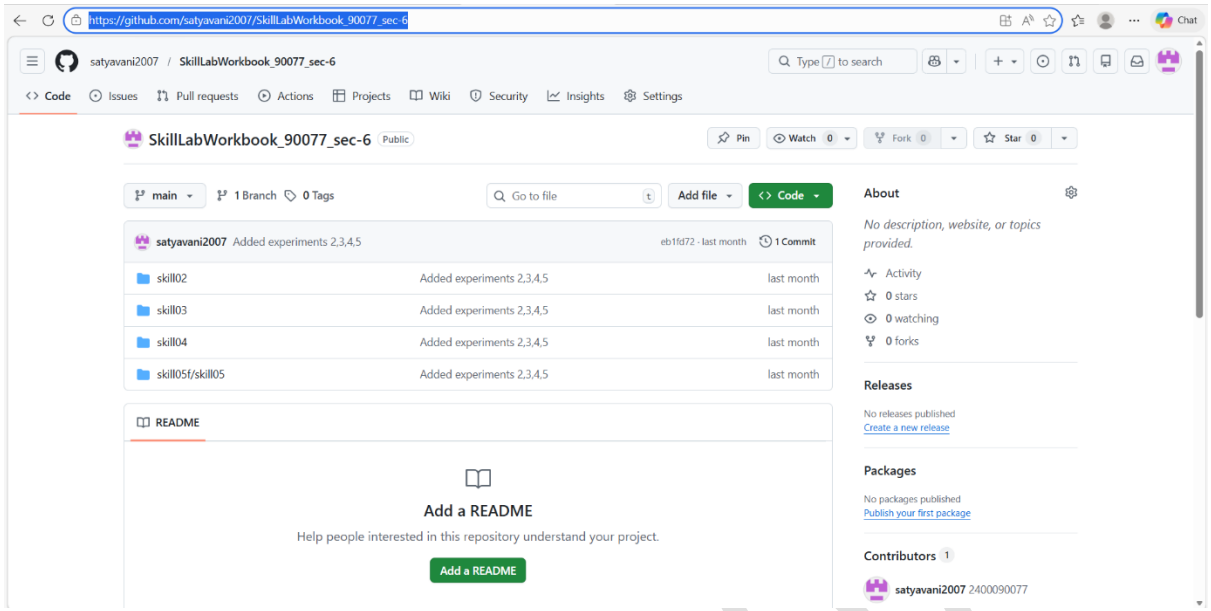
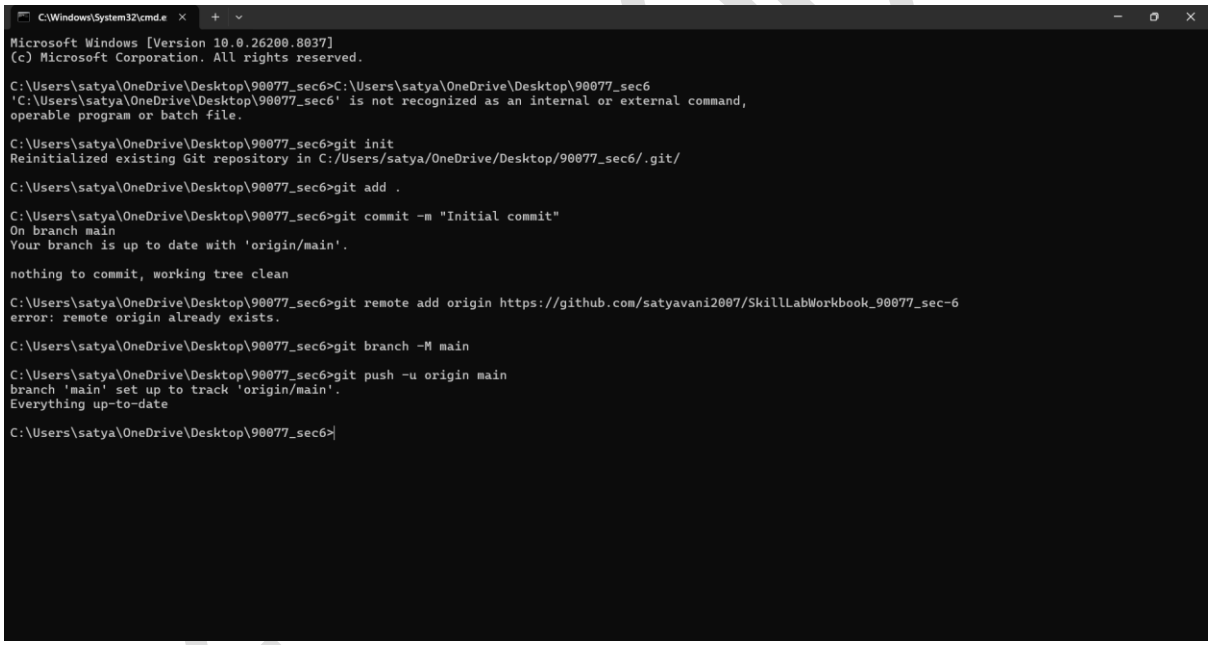
OUTPUT:





Git Hub Link:

[satyavani2007/SkillLabWorkbook_90077_sec-6]